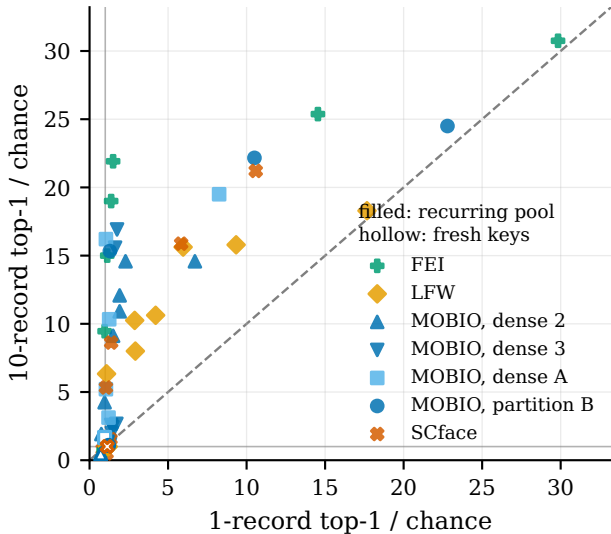
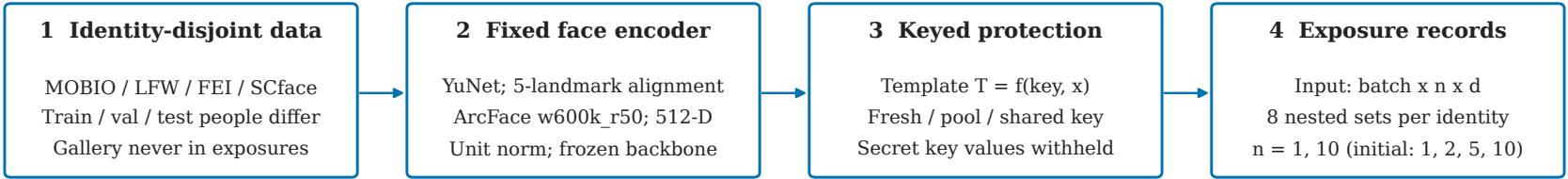


# BioHash record-count gains



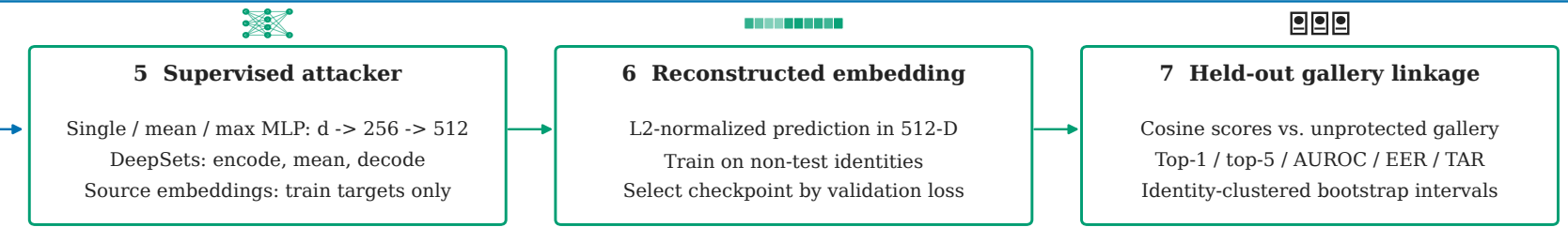
## A DATA AND PROTECTION



**OUTPUT d** BioHash: 128 bits | MLP-Hash: 512 bits | IoM-GRP: 300 codes ( $q=16$ ; one-hot  $d=4,800$ )  
PolyProtect: 170 reals (window  $m=5$ , overlap=2). Scheme coverage differs by dataset.

## B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



**KEY DESIGN** Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

**REPLICATION** MOBIO / FEI follow-up: 2 identity partitions x 3 model seeds; matched 120-epoch training caps.

**DESIGN** Same-identity aggregation under controlled transform reuse; detailed attacker paths in the companion figure.

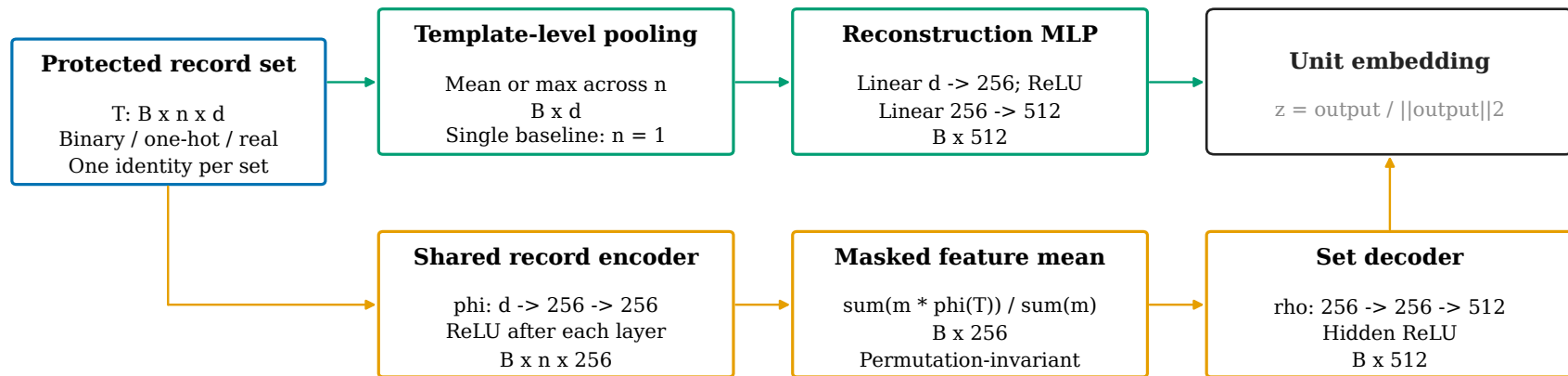
## A MATCHED EXPOSURE CONSTRUCTION

One identity, n source records; nested subsets from one fixed permutation per repeat; gallery excluded.

Fresh: independent source-record keys. Pool K: hidden transforms recur across identity splits. No keys or slots in model input.

**T:  $B \times n \times d$      $d = 128$  BioHash; 512 MLP-Hash; 4,800 one-hot IoM-GRP; 170 PolyProtect**

## B TWO AGGREGATION PATHS



Primary: mean-pool MLP. Secondary: DeepSets. m masks valid records; all n records are valid in these runs.

## C SUPERVISION

Target  $u = \text{normalize}(\text{mean of exposed source embeddings})$ .

Loss =  $\text{mean}[1 - \cos(\text{angle}(z, u))] + 0.1 \times \text{MSE}(z, u)$ .

Adam: lr 0.001; weight decay 0.0001; hidden width 256.

Train identities only; select minimum validation loss; freeze test.

B: batch size; n: exposure count; d: template input width; N: test gallery identities. Source targets are not inference inputs.

## D HELD-OUT IDENTITY LINKAGE

Scores =  $z \cdot G^T$ ;  $G: N \times 512$  unprotected unit gallery.

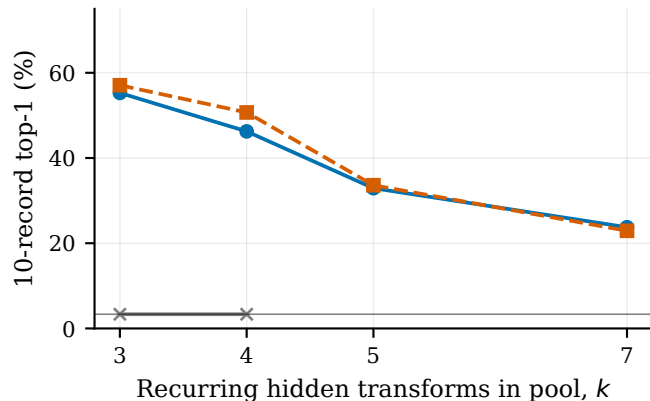
Rank identities; top-1/top-5, AUROC, EER, TAR at FAR.

Follow-up: 3 model seeds x 2 identity partitions; 120 epochs.

Crossed seed/identity intervals; paired primary contrasts.

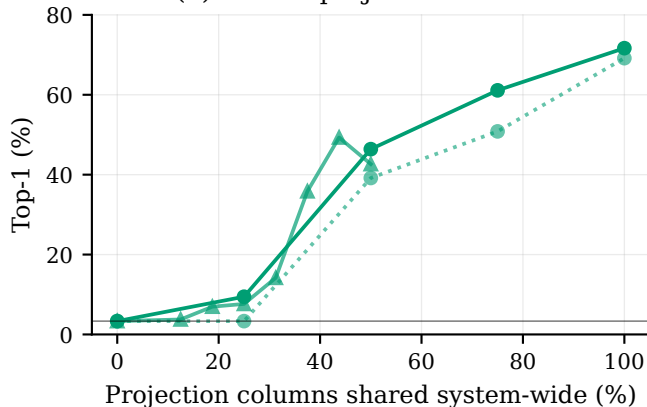
# MOBIO mechanism controls (30 gallery identities; chance 3.33%)

(a) Slot label vs. shuffled records



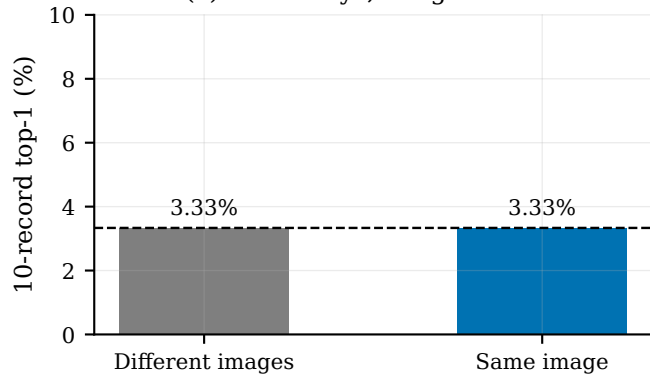
- slot hidden (DeepSets)
- slot label given to attacker
- records 2-10 from other identities

(b) Shared projection columns

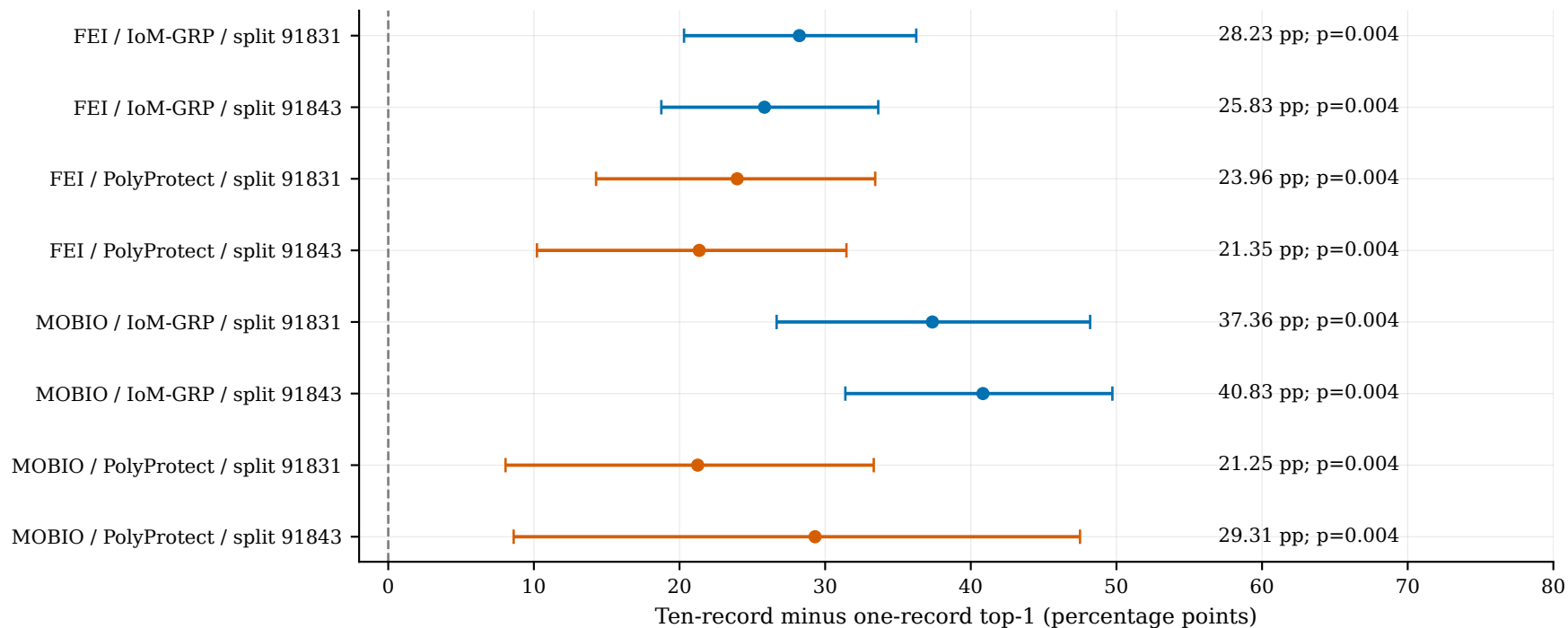


- 10 records, coarse sweep
- 10 records, fine sweep (new split)
- 1 record, coarse

(c) Fresh keys, image control



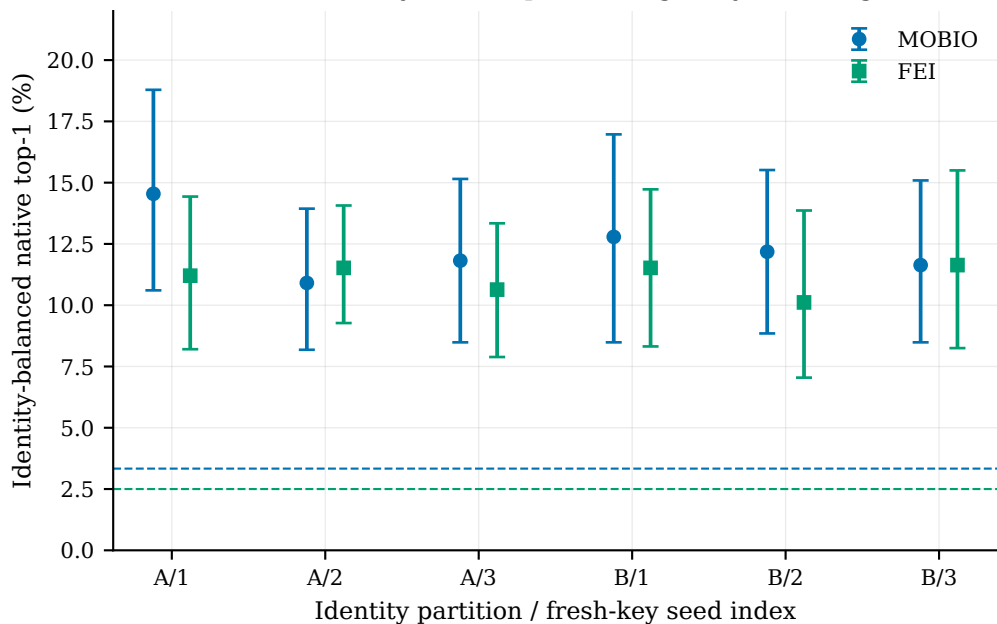
Pool-4 mean-pool amplification: three seeds, two identity partitions



Bars: crossed model-seed / identity bootstrap 95% intervals. Tests: identity sign-flips, Holm family of 8.

Matched 120-epoch caps; key/set seeds fixed. Partitions are separate sensitivity analyses, not independent replications.

(a) Fresh PolyProtect: protected-gallery matching

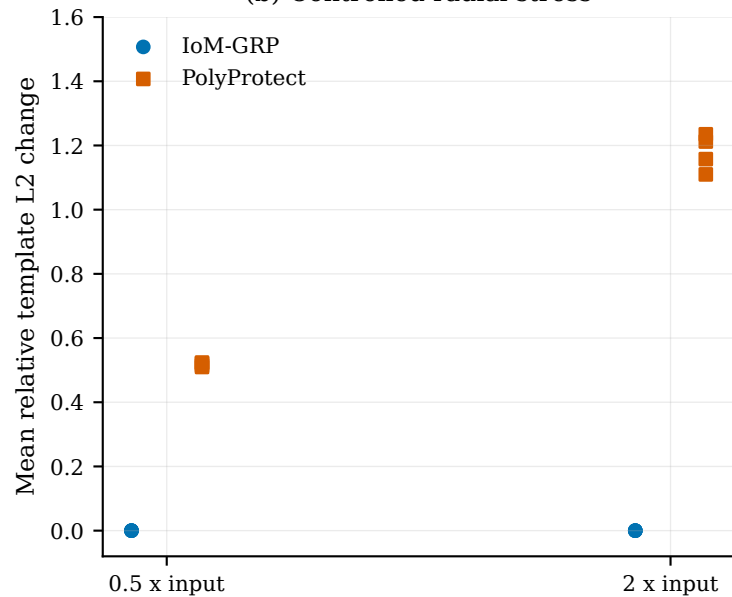


Native tests: 12 / 12 Holm-adjusted  $p \leq 0.006$ ; identity-clustered 95% intervals; dashed lines: chance.

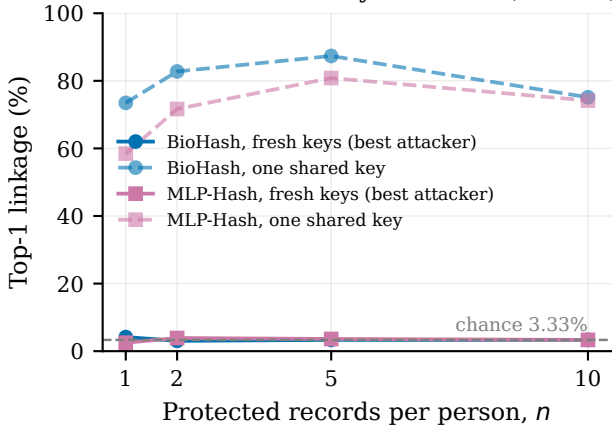
A/B: splits 91831/91843. Key seeds 1/2/3: 91873/91879/91883. Gallery-label permutations preserve probe clusters.

Radial stress: 32 held-out records per dataset/partition, fixed keys; measures scale sensitivity, not natural norm leakage.

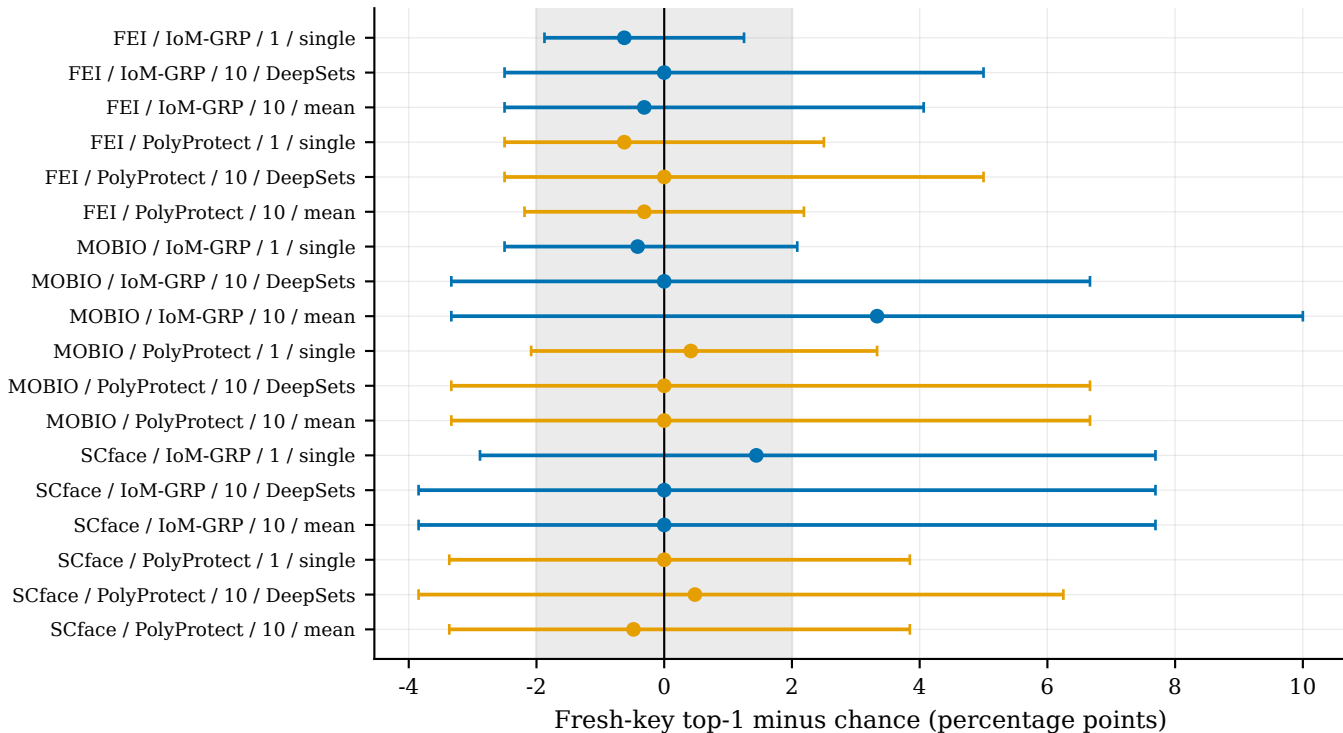
(b) Controlled radial stress



# Fresh and shared keys: MOBIO (N = 30)

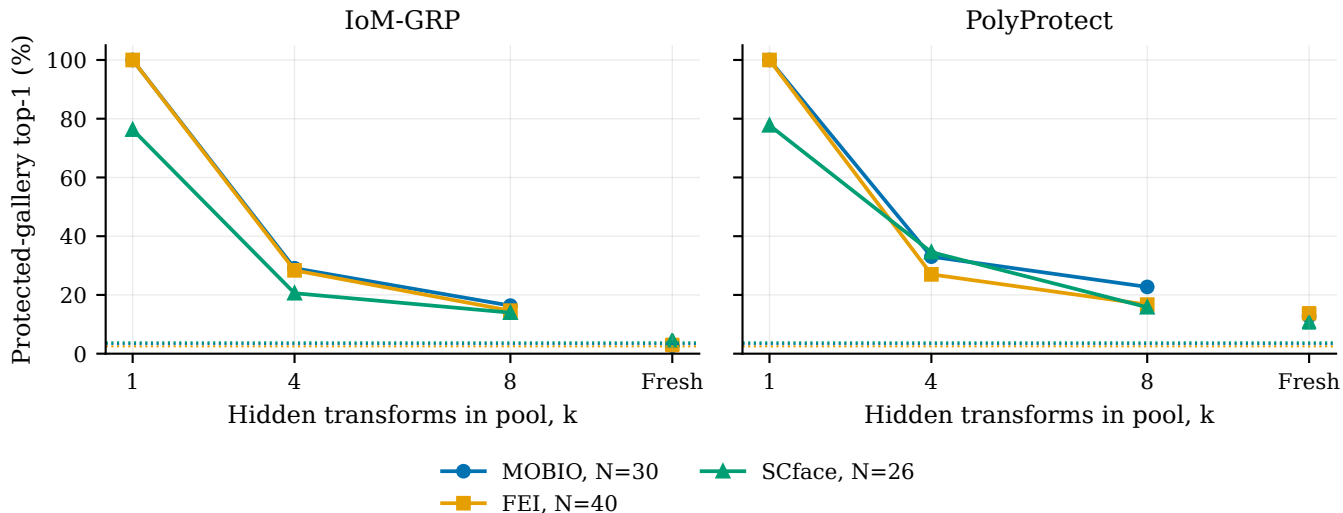


## Fresh-key 90% intervals: exploratory pilot sensitivity

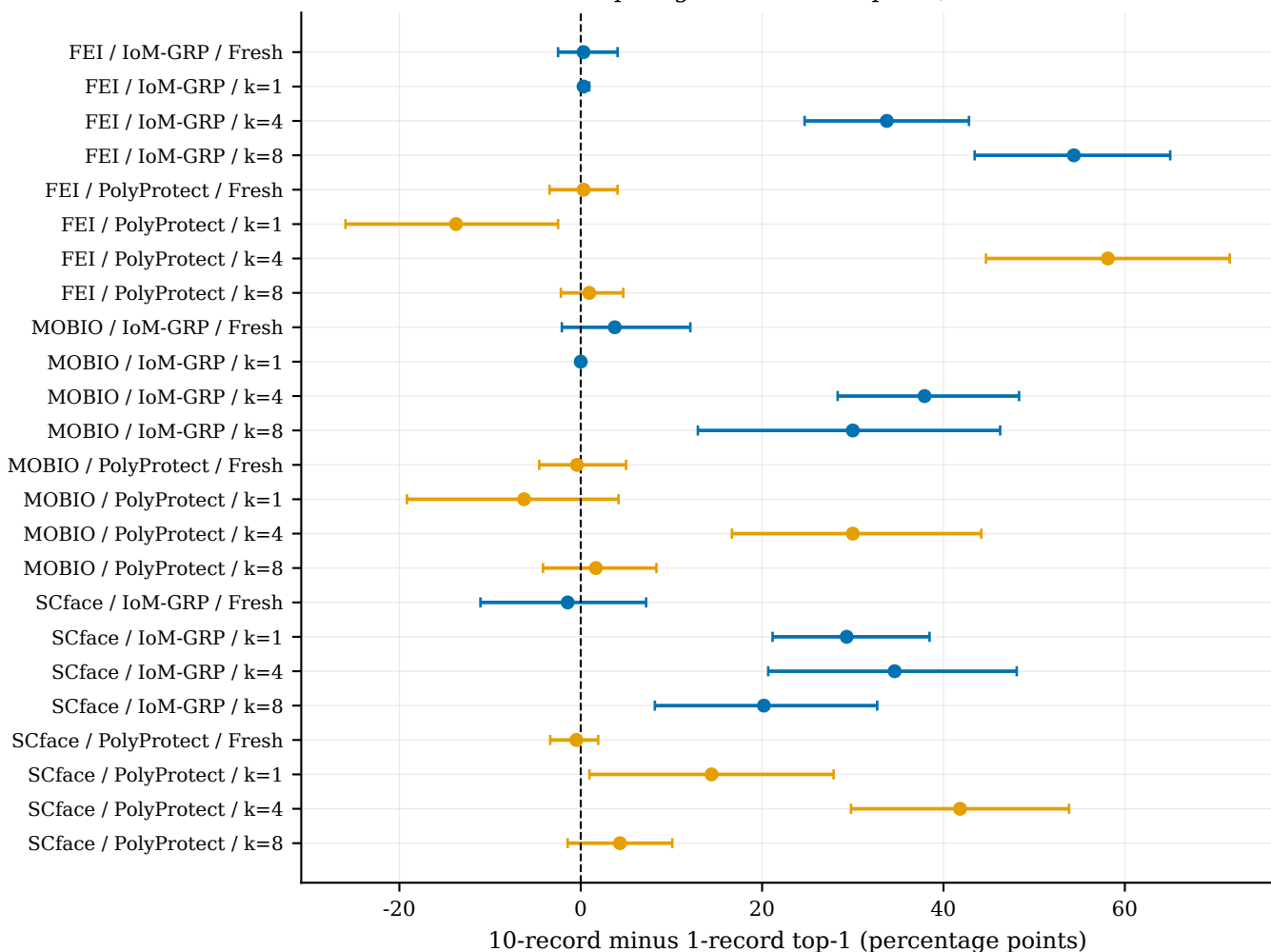


Illustrative +/-2 pp band; not an approved equivalence margin

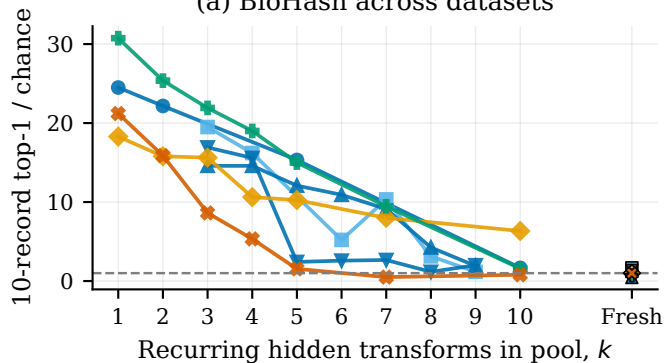
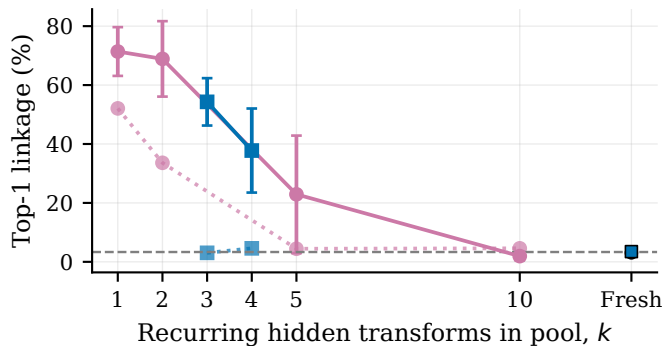
# Native matching diagnostic, not learned embedding linkage



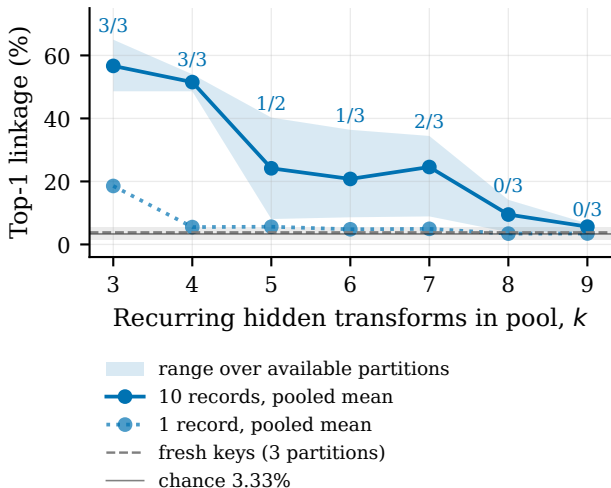
Paired mean-pool gains: one-seed pilots, 95% intervals



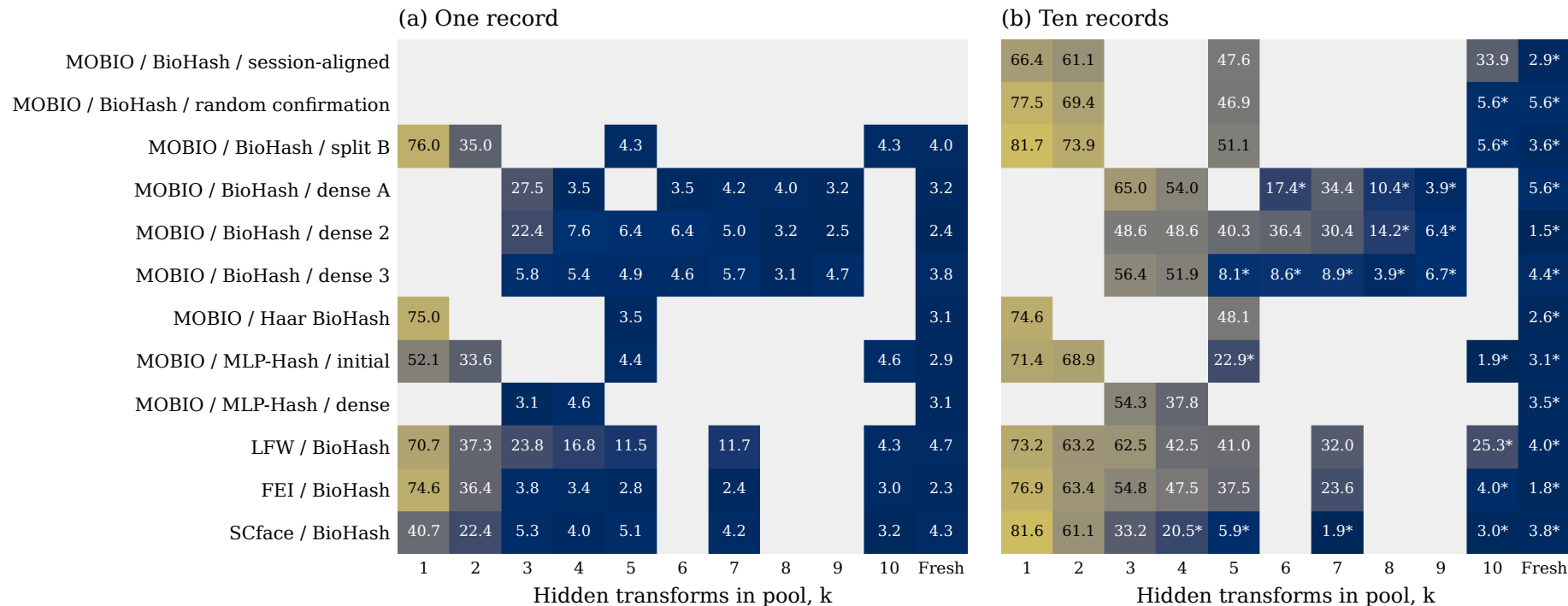
(a) BioHash across datasets

(b) MLP-Hash, MOBIO ( $N = 30$ )

# MOBIO boundary (N = 30)



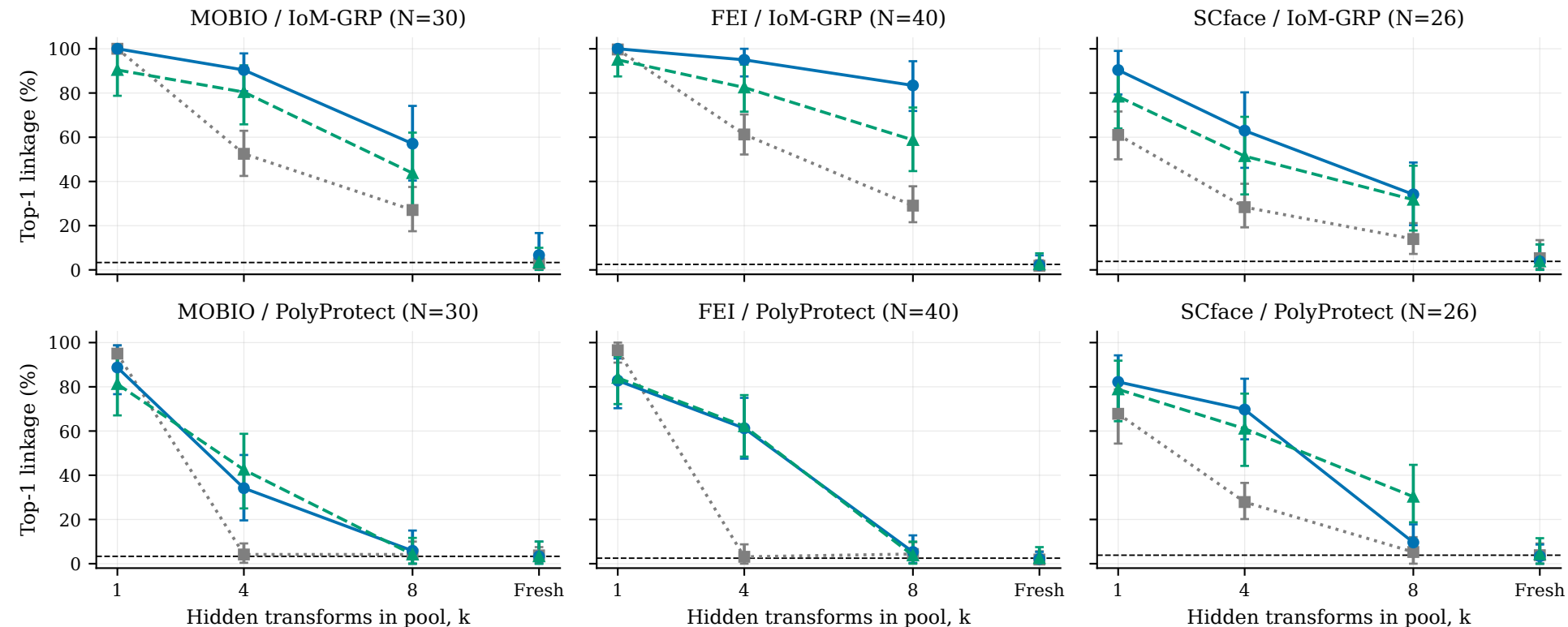
# Earlier three-seed key-pool studies



Cells: top-1 (%), 3-seed means. Grey: unavailable. \*: not all seed intervals exclude chance.  
 Gallery sizes: MOBIO 30; LFW 25; FEI 40; SCface 26. Rows are separate studies, not matched replications.

## New protection schemes: one-seed pilots

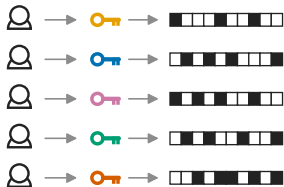
120-epoch cap; bars: identity-clustered 95% intervals; dashed baseline: chance



1 record 10 records, mean 10 records, DeepSets

### Fresh keys

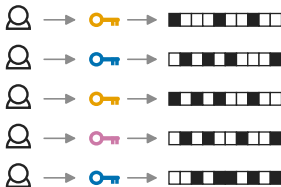
one key per source record



*no transform reuse*

### Recurring key pool

k transforms; slot labels hidden

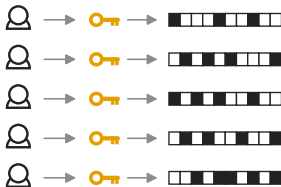


same colour = same transform

*reuse across train and test*

### Single shared key

k = 1; key value still hidden



*shared-transform baseline*

**Learned attacks inconclusive**

PolyProtect outside the theorem

**Multi-record gains observed**

boundary depends on dataset and split

**Strong linkage observed**

hidden does not mean unlearnable

### Protocol

Disjoint train / validation / test identities; one held-out gallery image per identity  
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals